

Unit: Unit 6: Systems of Linear Equations and Inequalities

Name: _____

Topic: Applications of Systems (Word Problems)

Date: _____

Advanced (Habanero)

Open Ended Questions (FRQ)

- Q9.** A bookshelf has 5 shelves, and each shelf can hold 8 books. If the bookshelf is currently empty, and x books are added to it, the number of books on the bookshelf is x . If 20 more books are added, the total number of books on the bookshelf is $x + 20$. How many books are on the bookshelf if it is currently $\frac{3}{4}$ full? Show your work and explain your answer.
- Q10.** A bakery sells a total of 500 loaves of bread per day. They sell a combination of whole wheat and white bread. If they sell x loaves of whole wheat bread, they sell $500 - x$ loaves of white bread. If the profit from whole wheat bread is 0.50 per loaf and the profit from white bread is 0.75 per loaf, what is the profit if they sell 250 loaves of whole wheat bread? Show your work and explain your answer.

Chef's Tips

- Make sure to read each question carefully.
- Use the given formulas and equations to solve the problems.
- Show all work and explain answers for free response questions.